

CAFA Integrated Program Management Information System (PMIS)

System Architecture & Functional Proposal

1. System Purpose

The CAFA Program Management Information System (PMIS) is designed to centralize, standardize, and strengthen all program management processes across CAFA headquarters and state offices.

The system will support planning, implementation, monitoring, reporting, coordination, beneficiary tracking, risk management, and project budget tracking across all operational states.

The platform is designed specifically for program operations only and does not include logistics, procurement, warehouse management, fleet management, HR management, payroll, or accounting ERP functions.

2. Core System Structure

The system architecture is based on interconnected modules where every operational record is linked through one centralized structure:

State

- Project
- Output
- Activity
- Indicator
- Beneficiary
- Budget
- Report
- Risk

Every activity, report, beneficiary, indicator, and budget line must be linked to a specific project and state.

3. Geographic Coverage

The system will support CAFA operations across 14 states.

Each state profile will include:

- State name
- Localities
- State office information
- Assigned State Office Manager
- Active projects
- Beneficiary statistics
- Implementation progress
- Reporting status
- Risk level
- Pending approvals
- Project performance summary

4. System Approach

The platform will operate as a centralized collaborative program management system connecting headquarters and field offices in real time.

The system will support:

- Multi-user collaboration
- Role-based permissions
- Approval workflows
- Real-time dashboards
- Automated reporting
- Evidence uploads
- Multi-state operations
- Arabic and English interfaces
- Future offline synchronization

5. User Roles and Permissions

The system will use Role-Based Access Control (RBAC).

Each user will have:

- Assigned role
- Assigned state
- Assigned projects
- Assigned permissions

Access will be restricted according to organizational responsibilities.

6. Headquarters Roles

6.1 Super Admin

Responsible for:

- Full system administration
- User management
- Role configuration
- System settings
- Permissions control
- Database administration

6.2 Executive Director

Responsible for:

- Strategic visibility
- High-level dashboards
- Organization-wide project overview
- Donor portfolio visibility
- Risk visibility
- Final strategic oversight

This role has read-only strategic access.

6.3 Program Manager

Responsible for:

- Full program oversight
- Project supervision
- Approval workflows
- Final project approval
- Final report approval
- Program performance monitoring
- Coordination between HQ and states

This role holds final organizational approval authority.

6.4 Senior Program Coordinator

Responsible for:

- Coordination review
- Workflow supervision
- Reviewing project consistency
- Reviewing reporting quality
- Following implementation progress
- Escalating operational issues

6.5 Technical Coordinators

Technical Coordinators are sector-based users responsible for technical quality assurance.

Possible sectors:

- WASH
- Health
- Protection
- Education
- Livelihoods
- Child Protection
- GBV
- MPCA

Responsibilities include:

- Technical review of projects
- Technical review of reports
- Indicator validation

- Activity review
- Technical comments
- Technical approval
- Alignment with donor standards
- Sector quality assurance

Technical Coordinators provide technical approval only.

They do not provide final organizational approval.

6.6 Budget Controller

This role is responsible only for project budget tracking.

This is not a finance or accounting role.

Responsibilities include:

- Budget tracking
- Burn rate monitoring
- Budget vs actual analysis
- Activity-level budget monitoring
- Budget alerts
- Remaining balance tracking
- Donor grant monitoring
- Liquidation tracking

This role does not manage:

- Accounting
- Payroll
- Banking
- Financial ERP systems

7. State-Level Roles

7.1 State Office Manager

Responsible for:

- State-level coordination
- Supervising implementation
- Reviewing reports
- Reviewing operational progress
- Following delayed activities
- Escalating risks and challenges
- Monitoring overall state performance

Access is restricted to assigned state only.

7.2 State Program Officer

Responsible for:

- Project implementation
- Activity management
- Field reporting
- Beneficiary updates
- Evidence uploads
- Progress tracking
- Creating project drafts
- Updating achievements

This role is the main operational program role at state level.

7.3 Field Officer

Responsible for:

- Field-level reporting
- Data entry
- Field monitoring
- Supporting implementation
- Uploading field evidence

7.4 Community Mobilizer

Responsible for:

- Beneficiary registration
- Community engagement
- Supporting field activities
- Attendance tracking
- Community data collection

8. Integrated Monitoring and Reporting Functions

The system does not use a standalone MEAL Officer role.

Instead, monitoring, evaluation, accountability, and learning functions are integrated across all operational roles.

Examples:

- Technical Coordinators manage indicators and technical quality.
- State Program Officers update field achievements.
- Program management reviews dashboards and reports.
- Management monitors performance analytics.

This creates a distributed monitoring structure integrated into the operational workflow.

9. Project Creation Workflow

Projects can be initiated from:

- State offices
- Headquarters technical teams

9.1 State-Level Project Initiation

State Program Officers can:

- Create project drafts
- Submit needs assessments
- Define target locations

- Estimate beneficiaries
- Propose activities
- Upload assessments

Project status:
Draft

9.2 State Review

The State Office Manager reviews:

- Operational feasibility
- Local priorities
- State readiness
- Accuracy of information

Project status:
State Reviewed

9.3 Technical Review

Technical Coordinators review:

- Technical quality
- Indicators
- Outputs
- Methodologies
- Sector standards
- Donor alignment

Project status:
Technically Approved

9.4 Coordination Review

Senior Program Coordinator reviews:

- Overall consistency
- Cross-sector integration

- Project readiness

Project status:

Coordination Approved

9.5 Final Approval

Program Manager provides:

- Final organizational approval
- Project activation

Project status:

Approved / Active

10. Report Approval Workflow

Reports follow the same multi-level approval process:

Draft

→ State Review

→ Technical Review

→ Coordination Review

→ Final Approval

The system must support:

- Approval comments
- Revision requests
- Rejection workflow
- Approval timestamps
- Approval history logs

11. Project Management Module

Each project will contain:

- Project code

- Project title
- Donor information
- Agreement reference
- Start date
- End date
- Target states
- Target localities
- Sector
- Outputs
- Activities
- Indicators
- Targets
- Budget lines
- Assigned staff
- Risks
- Reports
- Attachments

12. Beneficiary Management Module

The system will support integrated beneficiary management linked to:

- States
- Projects
- Activities
- Indicators
- Reports

Beneficiary categories may include:

- IDPs
- Returnees
- Host communities
- Women
- Children
- Farmers
- Schools
- Health facilities
- Community groups

Each beneficiary record may include:

- Beneficiary ID

- Name/code
- Age group
- Gender
- Vulnerability category
- State
- Locality
- Assistance received
- Date of assistance
- Verification status
- Supporting documents

13. Budget Tracking Module

The system includes project budget tracking only.

It is not a full financial accounting system.

The module structure:

Project Budget

→ Output Budget

→ Activity Budget

→ Actual Expense

→ Remaining Balance

Features include:

- Budget vs actual
- Burn rate
- Remaining balance
- Budget alerts
- Donor budget tracking
- Activity-level budget monitoring
- Liquidation status

14. Risk Management Module

Risks are linked to:

- States

- Projects
- Activities
- Reports

Risk categories include:

- Security risks
- Operational risks
- Access constraints
- Protection risks
- Compliance risks
- Financial misuse risks
- Data quality risks

Risk workflow:

Risk Identified

→ Assigned

→ Mitigation Plan

→ Follow-up

→ Escalation

→ Closed

15. Dashboards and Analytics

The system automatically generates dashboards from operational data.

Dashboards include:

- Active projects
- State performance
- Beneficiary achievements
- Budget utilization
- Burn rate
- Delayed activities
- Pending approvals
- High-risk states
- Sector performance
- Donor portfolio overview

16. Reporting System

The system generates:

- Activity reports
- Monthly state reports
- Project progress reports
- Donor reports
- Beneficiary reports
- Budget reports
- Risk reports
- Dashboard summaries

Reports should support:

- PDF export
- Excel export
- Auto-generated charts
- Real-time data integration

17. Audit Trail and Accountability

Every system action must be logged automatically.

Audit logs include:

- User name
- Action performed
- Module affected
- Old value
- New value
- Date and time
- Approval status

This supports:

- Accountability
- Transparency
- Donor compliance
- Internal control

18. Recommended Technical Stack

Frontend

- Next.js

Backend

- Node.js / NestJS or Django

Database

- PostgreSQL

Authentication

- JWT + RBAC

Storage

- Supabase Storage or Cloud Storage

Mobile Application

- Flutter (future phase)

19. System Language

The platform must support:

- English
- Arabic
- RTL layout support
- Multi-language switching

20. Future Mobile Features

Future mobile app features include:

- Offline data collection
 - Beneficiary registration
 - Field reporting
 - Evidence upload
 - Offline synchronization
 - Push notifications
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21. Implementation Phases

Phase 1

- User management
 - Roles and permissions
 - States module
 - Projects module
 - Dashboard setup
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Phase 2

- Activities
 - Indicators
 - Reports
 - Approval workflows
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Phase 3

- Beneficiary tracking
 - Risks module
 - Budget tracking integration
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Phase 4

- Advanced dashboards
- Analytics
- Donor dashboards

- Performance visualization
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Phase 5

- Mobile application
 - Offline synchronization
 - Field deployment tools
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22. Final System Vision

The CAFA PMIS is designed to become a centralized, collaborative, and scalable program management platform connecting headquarters and field offices across all operational states.

The system aims to:

- Improve coordination
- Strengthen accountability
- Improve reporting quality
- Improve visibility
- Support evidence-based decision making
- Standardize workflows
- Improve donor compliance
- Enhance operational efficiency across all program sectors and states.